Private Learning through the lens of Optimal Transport

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StatMathAppli 2019, Fréjus

What is privacy ?

Different notions of privacy

- **Differential Privacy:** cannot retrieve info from databases. Useful for sensitive data.
- **Economic/Rational Inattention literature:** info belief unchanged. Economic application.

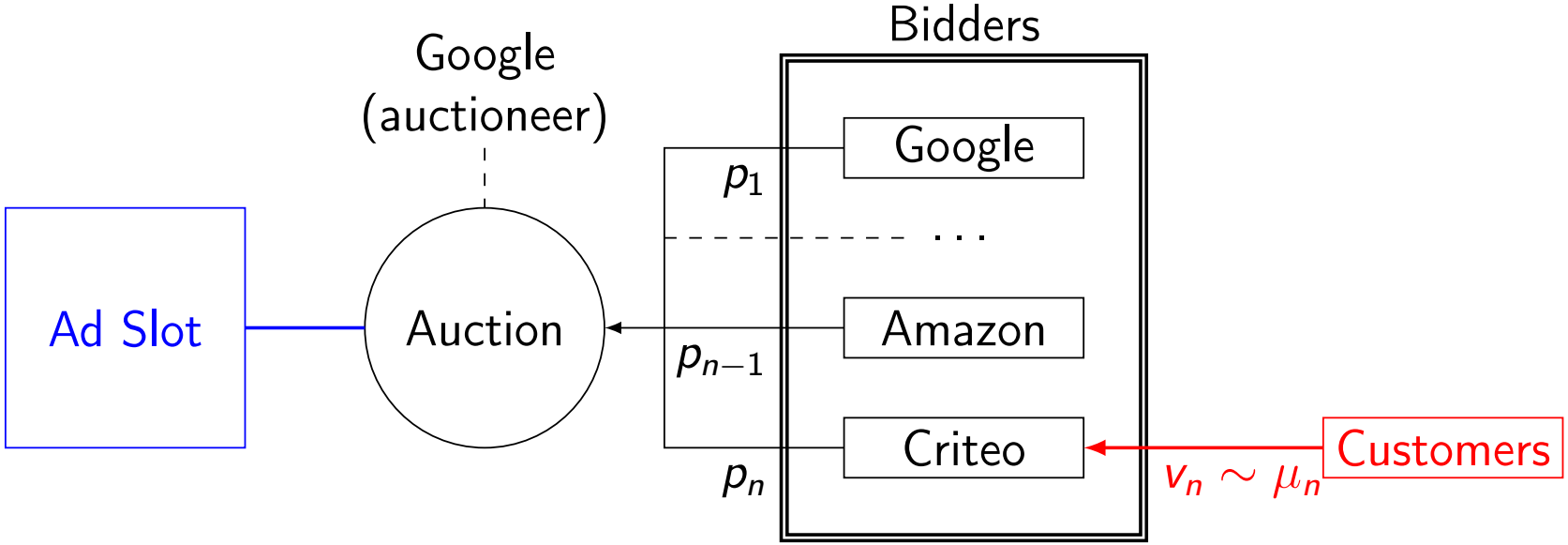
$\rightarrow$ Privacy affects utility.
Economic info is an economic resource.
Find best privacy and revenue trade-off ?

Some Applications

An economic example

Online repeated auctions

Ad slot valued v. Bid $p \implies$ auctioneer infers v.
Auctioneer's revenue $\nearrow$ while bidder's utility $\searrow$ when v public.



Online advertisement auction system

Bidder: short term utility **and** hide value distribution

Another example

Computation through external servers

Training deep NN

1. Send data to AWS
2. Train model on AWS
3. Get weights

$\rightarrow$ drawback: Amazon knows data.
Send "fake" data and keep accuracy ?

A Toy Example

Toy example

Player: minimizes utility loss

$$\min_{x \in \mathcal{X} \subset \mathbb{R}^d} x^\top y_k$$

y_k depends on **private type** $k \in \{1, \dots, K\}$ with prior $p_0 \in \Delta_K$.
Adversary: observes x and infers k

Toy example

Player: minimizes utility loss

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Adversary: observes x and infers k

$x \sim \mu_k \in \mathcal{P}(\mathcal{X})$ given $k \to$ posterior p_x given by Bayes rule:

$$p_x(k) = \frac{p_0(k)\mu_k(x)}{\sum_{j=1}^K p_0(j)\mu_j(x)}$$

Previous objective

Program in previous literature[1]:

$$\min_{\mu_1,\dots,\mu_K} \sum_{k=1}^{K} p_0(k)\ \mathbb{E}_{x\sim\mu_k}\left[x^\top y_k\right]$$
$$\text{such that } \mathbb{E}[KL(p_x, p_0)] \leq \varepsilon$$

[1] Eilat, R., Eliaz, K., and Mu, X. (2019). Optimal privacy-constrained mechanisms

Our objective

Our program:

$$\min_{\mu_1,\dots,\mu_K} \sum_{k=1}^{K} p_0(k)\ \mathbb{E}_{x\sim\mu_k}\left[x^\top y_k + \lambda\ KL(p_x, p_0)\right]$$

- Lagrange form
- Actual utility-privacy trade-off

$\rightarrow$ makes sense with privacy *for economic reasons*.

General model and some theorems

General formulation of the problem

General program:

$$\inf_{\substack{\gamma \in \mathcal{P}(\mathcal{X}\times\mathcal{Y}) \\ \pi_2 \# \gamma = p_0}} \int_{\mathcal{X}\times\mathcal{Y}} \left(c(x,y) + \lambda\, D(p_x, p_0)\right) \mathrm{d}\gamma(x,y) \quad \text{(P-OPT)}$$

- type $y \sim p_0 \in \mathcal{P}(\mathcal{Y})$
- $\pi_2 \# \gamma(A) = \gamma(\mathcal{X} \times A)$
- Toy example: $c(x,y) = x^\top y$ and $D = KL$

Convex for f-divergence

Definition (f-divergence)

D is an f-divergence if there is $f : \mathbb{R}_+^* \to \mathbb{R}$ convex, $f(1) = 0$ such that for all distributions P, Q: $D(P, Q) = \int f(\frac{\mathrm{d}P}{\mathrm{d}Q})\mathrm{d}Q$.

Examples: KL, reverse KL and TV.

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Theorem (Convexity)

If D is a f-divergence, then (P-OPT) *is convex in γ.*

(P-OPT) easy for finite $\mathcal{X}$ and $\mathcal{Y}$.

Finite support

Theorem (Finite prior support)

If $|\mathrm{supp}(p_0)| = K$*, for all* $\varepsilon > 0$*, we can look for a solution of* (P-OPT) *with support of size* $K(K+2)$*.*

(P-OPT) equivalent to

$$\inf_{\substack{\gamma \in \mathbb{R}_+^{(K+2)K} \\ x \in \mathcal{X}^{K+2}}} \sum_{i,k} \gamma_{i,k} c(x_i, y_k) + \lambda \mathbb{E}[\mathrm{D}(p_x, p_0)] \qquad \text{(F-OPT)}$$

$$\text{such that for all } j, \ \sum_{i=1}^{K+2} \gamma_{i,j} = p_0^j$$

$\rightarrow$ finite dimension ☺ but not jointly convex ☹

An optimal transport problem

Quick recall: Optimal Transport

Optimal transport between distributions μ and ν:

$$\inf_{\gamma \in \Pi(\mu,\nu)} \int c(x,y) \mathrm{d}\gamma(x,y)$$

with $\Pi(\mu,\nu) = \left\{\gamma \in \mathcal{P}(\mathcal{X} \times \mathcal{Y}) \mid \pi_1 \# \gamma = \mu \text{ and } \pi_2 \# \gamma = \nu\right\}$

- a tool to compare distributions
- OT distances widely used in recent applications
- heavy computational cost

Sinkhorn divergence minimization

Definition (Sinkhorn divergence)

$$\mathrm{OT}_{c,\lambda}(\mu,\nu) = \inf_{\gamma \in \Pi(\mu,\nu)} \int c \mathrm{d}\gamma + \lambda \int \log\left(\frac{\mathrm{d}\gamma}{\mathrm{d}\mu \mathrm{d}\nu}\right) \mathrm{d}\gamma$$

- entropic regularization $\implies$ fast OT distances approximation[2]
- $\rightarrow$ big data applications

[2] Cuturi, M. (2013). Sinkhorn distances: Lightspeed computation of optimal transport

Sinkhorn divergence minimization

New formulation of the problem

If D=KL, (P-OPT) equivalent to

$$\inf_{\mu \in \mathcal{P}(\mathcal{X})} \mathrm{OT}_{c,\lambda}(\mu, p_0)$$

and for finite support:

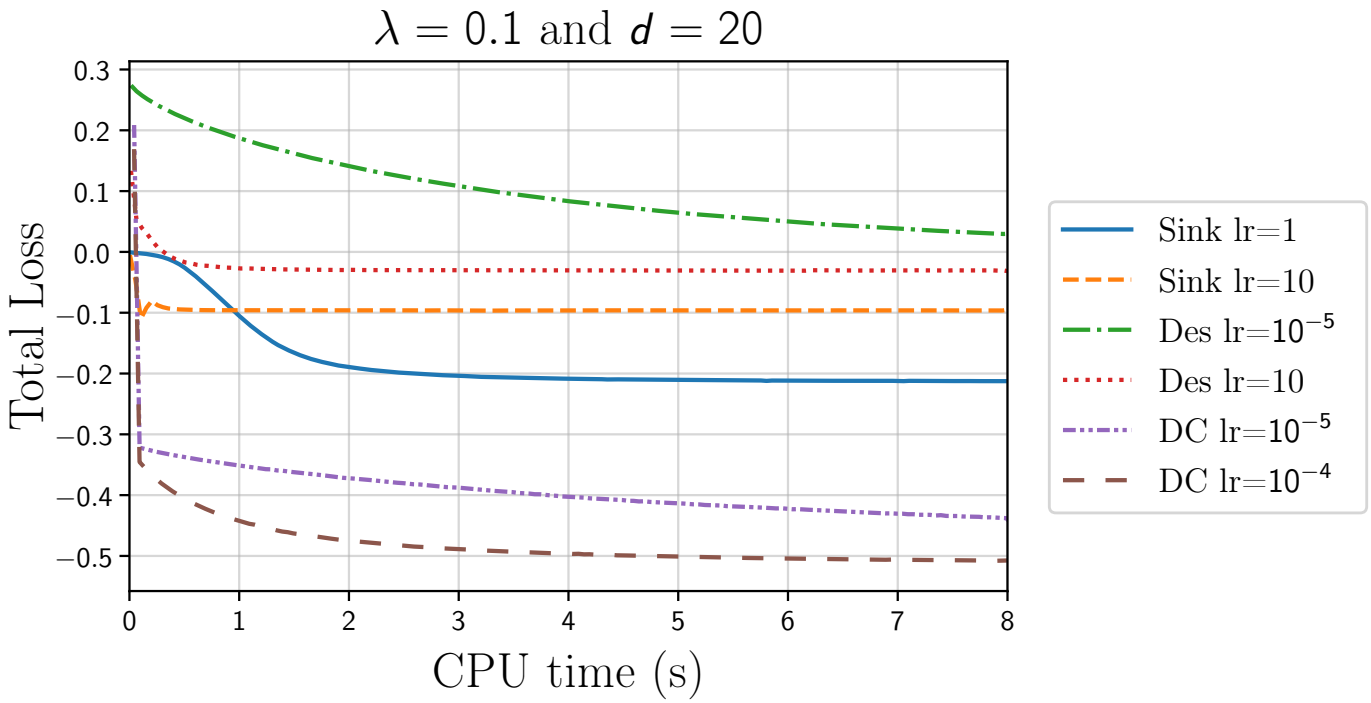
$$\min_{\substack{\alpha \in \Delta_{K+2} \\ x \in \mathcal{X}^{K+2}}} \mathrm{OT}_{c,\lambda}\Big(\sum_{i=1}^{K+2} \alpha_i \delta_{x_i}, p_0\Big) \qquad \text{(SINK-OPT)}$$

$\rightarrow$ lower dimension

Experiments

Expe 1: Toy example

$c(x, y) = x^\top y$, $\mathcal{X} = [-1, 1]^d$, $K = 100$ and D=KL.
DC: an even more adapted method for linear costs



Comparison of optimization schemes

- DC $\succ$ Sink $\succ$ Des
- Adaptation to problem structure is primordial

Expe 2: online repeated auctions

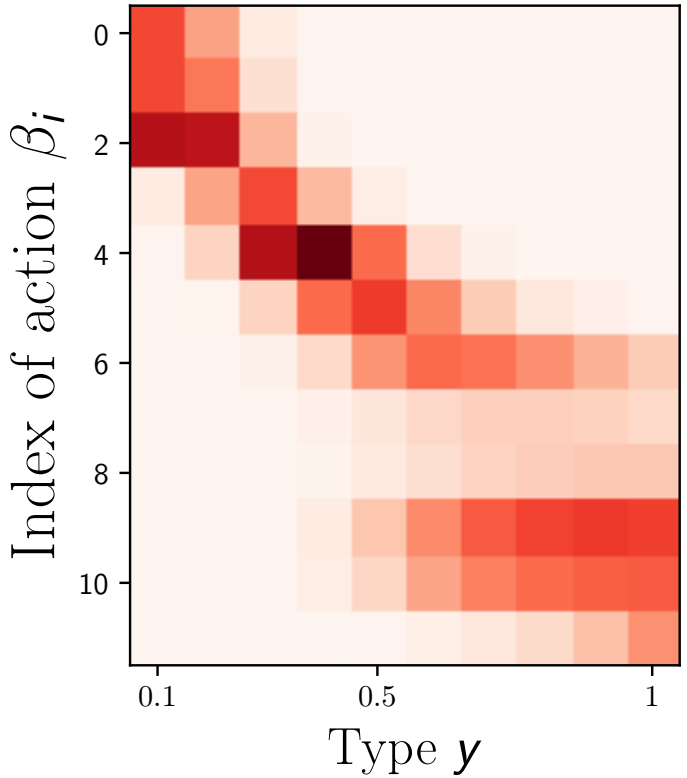
Problem description

- **Repeated auctions:** value $v \sim \mu_{y_j}$
- Fake distribution $\rho_i = \beta_i^j \# \mu_{y_j}$ by bidding $\beta_i^j(v)$
- If $\mu_{y_j} = \mathrm{Exp}(\frac{1}{y_j})$, then $\beta_i^j(v) = \beta_i(v/y_j)$
- $\mathcal{X}$ set of mappings $\mathbb{R}_+ \to \mathbb{R}_+$ (action β_i parametrized by a NN)
- Cost function for repeated auctions known[3]

[3] Nedelec, T., Karoui, N. E., and Perchet, V. (2019). Learning to bid in revenue-maximizing auctions

Expe 2: online repeated auctions

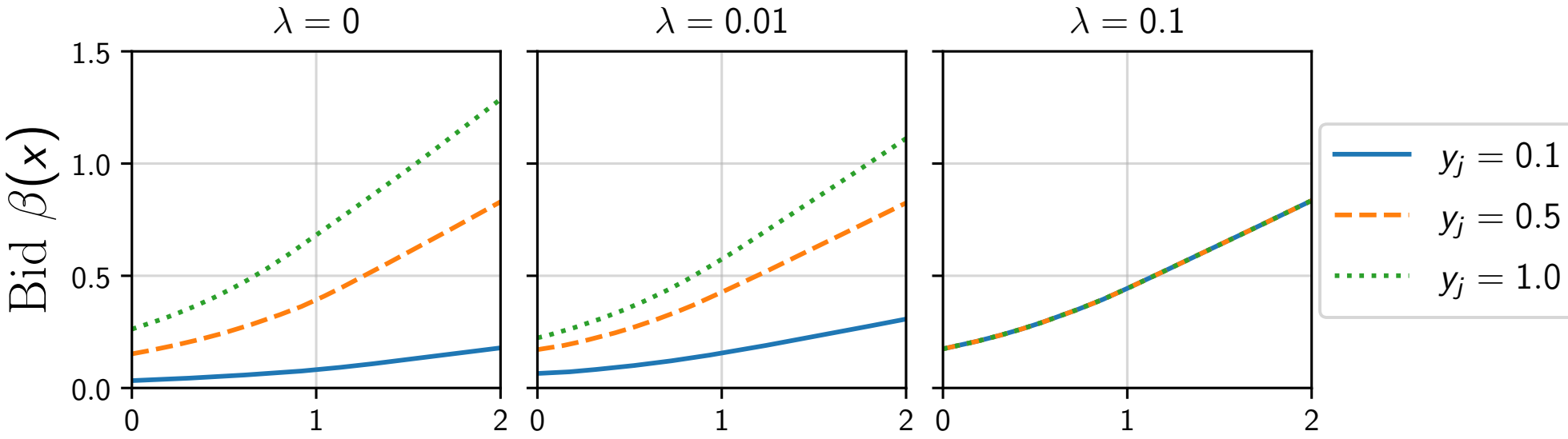
Partially revealing strategy



Joint distribution heat-map, with $\lambda = 0.01$

Expe 2: online repeated auctions

Strategy evolution with λ



Evolution of the most used β_i with the type and the regularization constant

Recap

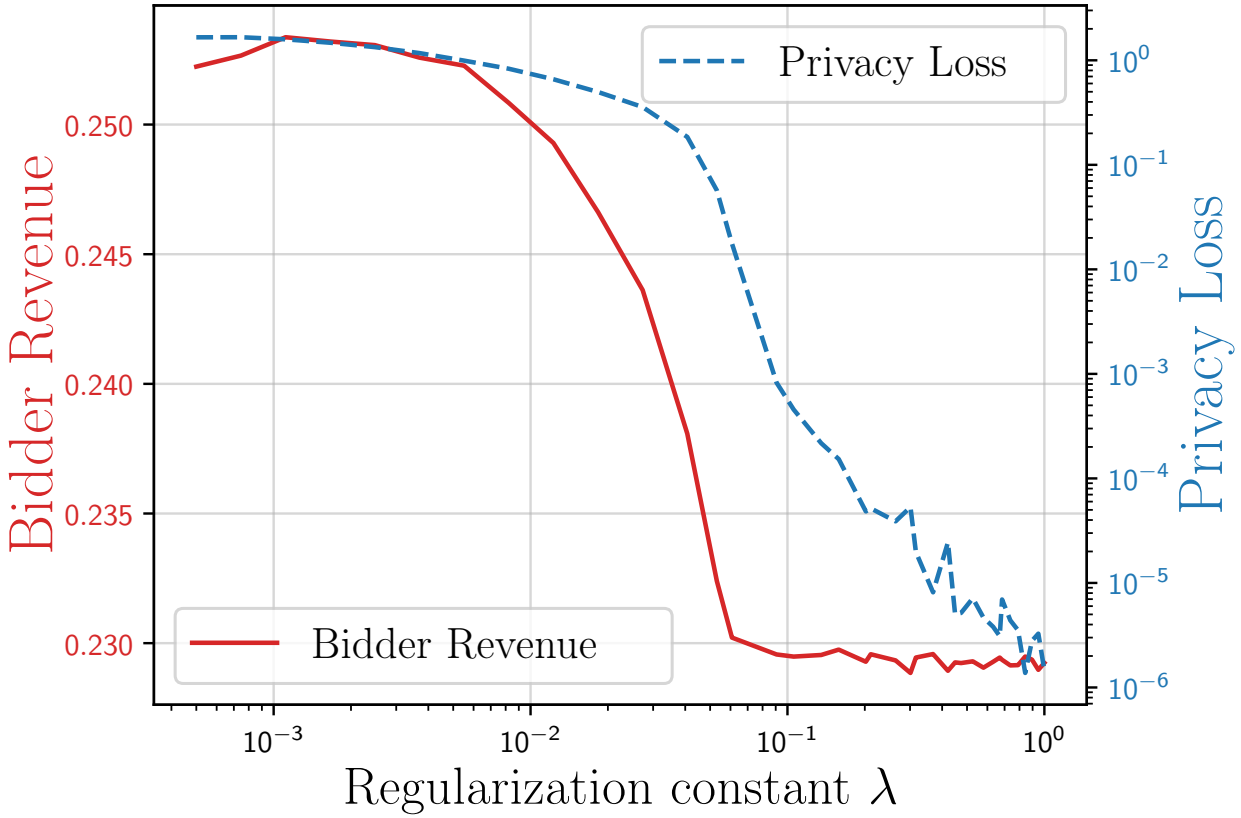
- utility-privacy trade-off motivated by economic mechanisms
- a regularized OT problem
- convexity + finiteness under mild assumptions
- benefit from Sinkhorn divergence / DC programming
- application for repeated auctions

Slides, code and paper at `eboursier.github.io`

Supplementary Slides

Expe 2: online repeated auctions

Utility-Privacy trade-off



Evolution of losses with λ

Linear loss $\implies$ DC program

DC program: $\inf_{x \in \mathcal{X}} f(x) - g(x)$ with convex f and g.

Theorem (DC program)

If $\mathcal{X}$ is an hyperrectangle and the loss is linear, (F-OPT) *is equivalent to the following DC program:*

$$\min_{\gamma \in \mathbb{R}^{(K+2)K}} \lambda \mathbb{E}[D(p_x, p_0)] - \sum_{i=1}^{K+2} \| \sum_{k=1}^{K} \gamma_{i,k} y_k \|_1$$

$$\textit{such that for all } k, \ \sum_{i=1}^{K+2} \gamma_{i,k} = p_0^k.$$

$\rightarrow$ specific algorithms such as DCA[4]

[4] Tao, P. D. and An, L. T. H. (1997). Convex analysis approach to dc programming: Theory, algorithms and applications

Skipped theorems

Theorem (Finite prior support)

If $|\mathrm{supp}(p_0)| = K$*, for all* $\varepsilon > 0$*, there exists an* ε*-optimal solution of* (P-OPT) *with support of size at most* $K(K+2)$*.*
Also, if $\mathcal{X}$ *is compact and* $c(\cdot, y)$ *is l.s.c. for every* y*, this also holds for* $\varepsilon = 0$*.*